PEDIATRIC *C. DIFFICILE*

7/2021

General Risk Factors

- Antibiotic use within 3 months
- Hospitalization within 30 days
- Acid suppressive therapy
- History of prematurity
- History of abdominal surgery

Risk Factors for SEVERE Disease

- Immunosuppressed state
- Severe comorbid disease(s)
- Hirschsprung's disease
- Dysmotility disorder
- IBD

Indications for Testing

Patients must be symptomatic to send testing

<12 months:
Rarely indicated; High rate of asymptomatic carriage

Symptomatic infants if no alternative etiology* w/:
- Risk factors for severe CDI
- History of frequent antibiotic exposures
- C. diff outbreaks in hospital unit

12-36 months:
Indicated if no alternative etiology* for diarrhea

See ">36 months" section for indications

>36 months:
Indicated if patient meets any of the following criteria:
- Diarrhea (≥3 unformed stools within 24 hours w/o alternative explanation)
- Unresolved symptoms after full treatment course
- Colitis on imaging w/accompanying symptoms
- Leukocytosis (WBC >15k cells/mm^3) & ileus
- Abdominal pain with radiographic evidence of bowel thickening
- Toxic megacolon
- Pseudomembranes

Testing Results

"*Clostridium difficile* by PCR" assay is the preferred test

PCR Result	Toxin Result	Interpretation
-		No toxigenic *C. difficile* present ~99% negative predictive value
+	+	**Toxigenic *C. difficile* present.**
+	-	May represent colonization or active clinical infection (requires clinical correlation to determine if treatment is warranted)

GI Panel is for use in patients w/community-acquired diarrhea (send in the first 3 days of hospitalization)
or
Hospitalized patients at risk for etiologies other than C. difficile (e.g., norovirus outbreak)

There is no role for performing both tests

Classifications

Mild/moderate
- CDI and no criteria in the "severe" or "complicated" categories

Severe

At least 2 abnormal lab values:
- WBC ≥15K
- SCr ≥1.5x baseline
- ANC ≤500
- ALB ≤2.5

OR

At least 1 high-risk condition:
- SOT/BMT <100 days
- Small bowel CDI
- IBD
- Hirschsprung's disease
- Dysmotility disorder
- Neutropenic from malignancy

Complicated

ANY of the following:
- Septic shock
- Severe sepsis
- Ileus or bowel obstruction
- Toxic megacolon
- Peritonitis
- Bowel perforation

Prophylaxis (can be considered for high risk patients)

Broad Spectrum Antibiotics

3 or more episodes in the past year
- **Vancomycin** PO 10 mg/kg/dose PO BID (max: 125 mg/dose)
  - Continue for the duration of broad spectrum antibiotic exposure

Treatment

Treatment is only indicated in **SYMPTOMATIC** patients
- Discontinue/change antibiotics if possible
- Avoid PPI/H2 blockers without an appropriate indication
- Implement infection control measures: Infection Control C. diff

1st episode

Mild/moderate
- **Metronidazole** 7.5 mg/kg/dose PO QID x10 days (max: 500 mg/dose)

Severe
- **Vancomycin** 10 mg/kg/dose PO QID x10 days (max: 125 mg/dose)

1st Recurrence*

***Recurrence defined as repeat infection within 8 weeks of previous episode**
- If metronidazole was used for 1st episode:
  - **Vancomycin** 10 mg/kg/dose PO QID x10 days (max: 125 mg/dose)
- If vancomycin was used for 1st episode:
  - **Vancomycin** 10 mg/kg/dose PO QID x14 days (max: 125 mg/dose) w/taper‡

2 or more recurrences*
- **Consult Pediatric Infectious Diseases**
  +
- **Vancomycin** 10 mg/kg/dose PO QID x14 days (max: 125 mg/dose) w/taper‡
  OR
- **Fidaxomicin** 16 mg/kg/dose PO BID x10 days (max: 200 mg/dose)
  OR
- **Fecal microbiota transplantation**

Triple therapy

Complicated
- **Consult Pediatric Infectious Diseases & Pediatric Surgery**
  +
- **Vancomycin** 10 mg/kg/dose PO QID x10 days (max: 500 mg/dose)
  +
- **Metronidazole** 7.5 mg/kg/dose IV q6h (max: 500 mg/dose)
  +
- **Vancomycin retention enema**§ 10-20 mL/kg/dose q6h (max: 500 mL/dose)

Follow-up

- Clear improvement should be seen after 3-5 days
- If failure to improve on PO metronidazole after 3-5 days, change to vancomycin PO
- If clinical deterioration occurs (worsening abdominal distention/pain, and/or peritonitis, worsening leukocytosis, end organ failure, intubation, vasopressor requirement, mental status changes, new or worsening acute kidney injury, or worsening lactate >5 mmol/L):
  - Consult Pediatric Infectious Diseases
  - Consult Pediatric Surgery
  - Change to triple therapy
  - Consider CT scan of abdomen/pelvis

Notes

*Possible alternative explanations of diarrhea:
- Laxative use within last 48 hours
- Tube feeds initiated in last 48 hours
- Oral contrast receipt in last 48 hours

‡**Vancomycin** taper (to be started after treatment course is completed) (max: 125 mg/dose):
- 10 mg/kg/dose PO BID x7 days
- 10 mg/kg/dose PO Daily x7 days
- 10 mg/kg/dose PO every other day x7 days
- 10 mg/kg/dose PO every 3 days x4 weeks

§Vancomycin retention enema administration
- 10-20 mL/kg/dose, max: 500 mL, of a 500 mg/500 mL solution in NS q6h instilled by appropriately sized Foley catheter inserted into rectum with balloon inflated and Foley clamped for 1 hour
- Treatment naïve patients should be started at 10 mL/kg/dose and escalated as tolerated up to 20 mL/kg/dose up to a max: 500 mL
- Suspected or known bowel perforation is a contraindication for rectal administration

Postoperative diverting loop ileostomy regimen consists of antegrade vancomycin flushes (500 mg in 500 mL of Lactated Ringers; q8h for a duration of 10 days) via a 24 French Malecot catheter in the efferent limb of the ileostomy and intravenous (IV) metronidazole (500 mg q8h) for 10 days. See Reference Neal MD, et al. Ann Surg 2011;254:423-7.

ALB: Albumin
ANC: Absolute neutrophil count
BMT: Bone marrow transplant
CDI: Clostridium difficile infection
GI: Gastrointestinal
IV: intravenously
PO: Orally
NS: Normal Saline
QID: Four times daily
q6h: Every 6 hours
SCr: Serum creatinine
SOT: Solid organ transplant
WBC: white blood count

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source.